

The tools available today are inexpensive too. For instance, there is software (EPrints, DSpace, etc) that is free. Help, if needed, to set up such repositories is readily available.

He argues: "Such repositories will facilitate e-science to transform into inclusive science from an activity, at present, of an elitist club of rich nations. They can also help transform the current 'safari' kind of research performed by advanced countries into true collaboration with poor countries, especially in areas such as drug development for combating diseases such as malaria and tuberculosis."

The push for preprints

Arun makes a case for preprints to replace journal articles "as the principal mode of knowledge exchange in science and scholarship." This basically means that authors who work on a paper willingly submit their work for sharing, in its draft form, so that the flow of knowledge does not get blocked.

In the world of academic publishing, a preprint refers to an early version of a scholarly or scientific paper that precedes formal peer review and publication in a scholarly or scientific journal. It can be shared by its author, often prior to typesetting, free and before it gets published in a journal. Today, there are many preprint repositories. A listing, from A to Z, is available on Wikipedia (https://en.wikipedia.org/wiki/List_of_preprint_repositories).

Arun points out that we have arXiv (with over 2 million preprints deposited over 30 years), bioRxiv, chemRxiv, and other such global preprint servers.

However, not enough is being done in India in this space, he feels. In the meanwhile, commercial publishing companies are playing a domineering role not only in deciding what is and is not important in science but also in evaluating individual researchers and ranking institutions.

Arun notes that, unfortunately, even after two decades post Harnad's call to

researchers worldwide to self-archive, many scientists in India do not "bother to self-archive" and share their work. Sometimes, an intermediary (like a librarian) has to do the archiving.

"In a world of inequality, the open access initiative seeks to provide people all over the world (irrespective of where they live) with equal access to knowledge and information. The developing world stands to gain from the open access initiative but there also are obstacles to its realisation in this world," says Arun. He points to the many recommendations that governments, educational and research institutions as well as organisations and groups in developed countries need to take up to attain the noble objectives of the open access initiative in the developing world.

Indian initiatives

So, how are Indian OA initiatives faring?

Some journals have been shifting over to open access. These include journals published by various councils and academies, and a few by private publishers too. Yet, many Indian researchers still publish in "toll-access journals." There's a workaround here, though. Researchers and scholars can make their post-prints available. Efforts are on to "persuade" them to do so.

For instance, the prestigious IISc (Indian Institute of Science) in Bengaluru has a repository. It was the first to be set up in India, and one of the earliest in the world. Yet, Arun complains, the faculty and students "don't seem to be taking it seriously."

Interestingly, although the DST, DBT and CSIR mandate that research funded by them should be published for open access, researchers don't take it seriously and these agencies don't enforce it. "Science policy is not just about tabling reports before committees; it is also about persuading decision-making bodies to adopt certain policies and take action," Arun argues.

Different models of open access

GREEN: Self-archiving by authors. An author can post work to a website controlled by self, an institution that is funded, or an independent repository. Readers can download without paying. If an author posts a near-final version of his/her work after peer review, the archived version is called a 'postprint'. It can be the accepted manuscript as returned by the journal to the author after successful peer review.

GOLD: Publisher makes all articles and content available for free immediately on the journal's website — articles licensed for sharing and reuse via Creative Commons licences or similar. This may follow an 'author-pays' model.

HYBRID: Mix of open access and closed access articles. The publisher is partially funded by subscriptions, and provides open access for articles where the author or research sponsor pays a publication fee. It costs more than gold and can offer a lower quality of service. 'Double dipping' is where both authors and subscribers are charged, and is considered a controversial model.

BRONZE: Free to read only on the publisher's page, but lacks a clearly identifiable licence. Usually, no reuse is allowed.

DIAMOND: Authors are not charged an article processing charge (APC). Often, the journal requires funding from external sources — advertisements, academic institutions, learned societies, philanthropists or government grants. This model is usually small and multilingual.

BLACK: Unauthorised digital copying to enable free access to paywalled literature.

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